

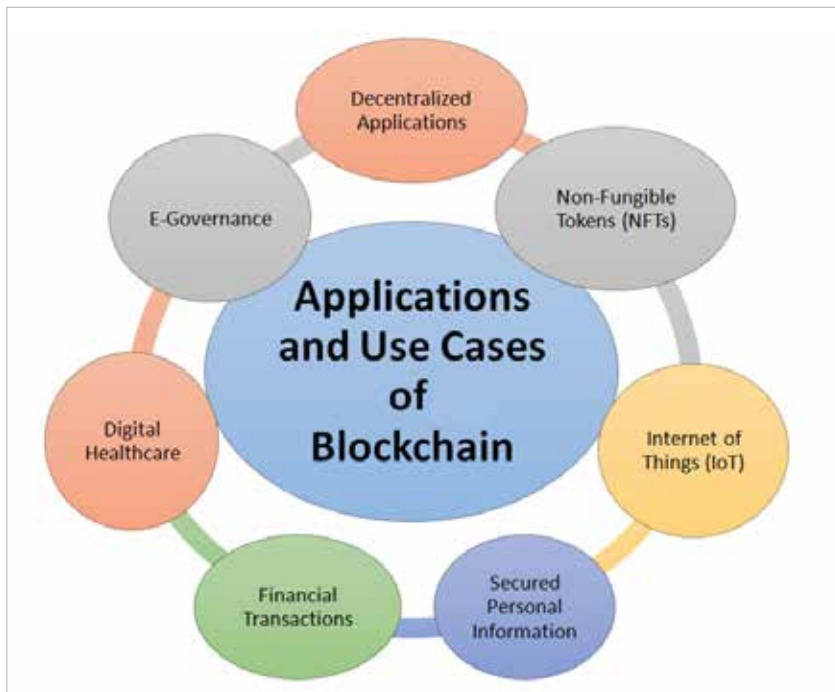


Figure 2: Applications and use cases of blockchain

makes use of blockchain protocols for storing data in a decentralised manner for secured transactions. DApps offer greater transparency, security and user control, making them a vital component of the developing Web3 ecosystem.

Decentralised finance (DeFi): One of the most popular use cases of DApps is in the realm of decentralised finance (DeFi) as it allows users to utilise peer-to-peer financial services without requiring banks or financial institutions. Uniswap, Aave and MakerDAO are some examples of such platforms that enable trading, lending and borrowing of digital assets without intermediaries like centralised exchanges or traditional banking systems.

These applications run on smart contracts that automate transactions, lowering operational costs and providing a global financial reach to the unbanked population. All transactions are recorded on the blockchain and are thus tamper-proof and auditable, providing transparency.

Supply chain management: Decentralised applications are

revolutionising supply chain management by enhancing transparency, traceability and efficiency. Blockchain is already used by companies such as IBM Food Trust and VeChain to monitor the supply chain movement from the manufacturer to the consumer. Using a decentralised ledger to log every step of the supply chain helps businesses prevent fraud, counterfeiting and inefficiencies. In the food industry, for example, blockchain-based supply chain DApps enable consumers to verify the authenticity and source of products, ensuring food safety and ethical sourcing. In pharmaceuticals as well, blockchain is being used to help tackle the challenge of counterfeit drugs as it helps to validate the genuineness of medicines before they are delivered to patients.

Entertainment sector, gaming and NFTs: Blockchain based gaming and NFTs have ushered a seismic renaissance in the gaming industry. DApps are powering video games like Axie Infinity, Decentraland and The Sandbox, allowing players to freely

own in-game assets, trade collectibles and monetise their gaming experiences. In contrast to traditional games where virtual items are controlled by centralised companies, blockchain gaming allows players to own digital assets in the form of NFTs. These non-fungible tokens are digital collectible assets tied to a single universe in virtual space where players are able to buy, sell and transfer them between platforms, creating a decentralised gaming economy.

In addition, NFT marketplaces like OpenSea and Rarible allow artists, musicians and other content creators to tokenise their works, which guarantees their authenticity.

Decentralised social media and content platforms: Blockchain DApps are solving issues around data privacy, censorship and revenue distribution in the social media and content creation sphere. Social networks like Mastodon, Steemit and Lens Protocol use Web3 technologies and offer users decentralised alternatives to traditional social networks, where they can interact, share content and, potentially, earn rewards without corporate control of their data.

Classical social platforms rely on monetising user data whereas decentralised social media allows users to control their data and content. Similarly, blockchain-powered content platforms (Audius or LBRY) allow musicians, writers and video creators to directly disseminate their work without needing intermediaries, ensuring each participant earns his or her share.

Healthcare and data security: Healthcare systems also benefit from blockchain-based DApps. Medicalchain and BurstIQ are blockchain health platforms that help patients as well as healthcare providers to securely store and share medical records with each other using cryptographic encryption. Using blockchain for healthcare prevents unauthorised access to sensitive information and